

Symmetries and classical mechanics

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$$\text{or, } 2T - T + V = \text{const.}$$

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$$\text{or, Total energy, } T + V = \text{const.}$$

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QM :

commutation of an operator with Hamiltonian is the requirement for conservation of that observable

$$\text{TDSE, } \hat{\mathcal{H}}\psi = i\hbar \frac{\partial \psi}{\partial t} \implies$$

$$\text{Heisenberg Eq. of motion : } \frac{d\langle \hat{\mathcal{O}} \rangle}{dt} = \frac{1}{i\hbar} \langle [\hat{\mathcal{O}}, \hat{\mathcal{H}}] \rangle + \left\langle \frac{\partial \hat{\mathcal{O}}}{\partial t} \right\rangle$$

For no explicit time-dependence in the operator

$$\frac{\partial \hat{\mathcal{O}}}{\partial t} = 0 \text{ and then } \frac{d\langle \hat{\mathcal{O}} \rangle}{dt} = 0 \implies [\hat{\mathcal{O}}, \hat{\mathcal{H}}] = 0$$

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If Hamiltonian has continuous translational symmetry, then it is unchanged under any translation, including an infinitesimal one; equivalently it commutes with the translation operator, and hence :

$$[\hat{H}, \hat{T}_\delta] = [\hat{H}, \hat{T}_\delta] = [\hat{H}, \hat{1} - \frac{i\delta x \hat{p}}{\hbar}] = 0 \implies [\hat{H}, \hat{p}] = 0 \implies$$

conservation of linear momentum.

similarly, define x component of angular momentum as an operator whose conservation follows from

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- holds also for half-integral j values where the spherical harmonics do not serve as basis functions

Unitary operators:

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operator $\hat{U}$ is unitary if $\hat{U}^\dagger = \hat{U}^{-1} \implies \hat{U}\hat{U}^\dagger = \hat{1} = \hat{U}^\dagger\hat{U}$

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or, $|\tilde{\phi}\rangle = \hat{\tilde{A}}|\tilde{\psi}\rangle$, where $\hat{\tilde{A}} = \hat{U}\hat{A}\hat{U}^\dagger$

under unitary transformation, inner products preserved but operator changes

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$$\hat{\tilde{H}} = \hat{\Pi} \hat{H} \hat{\Pi}, \text{ or } \hat{\Pi} \hat{H} = \hat{H} \hat{\Pi} \implies [\hat{H}, \hat{\Pi}] = 0 \implies$$

$\exists$ complete set of simultaneous eigenstates of $\hat{\Pi}$ and $\hat{H}$.

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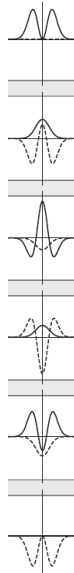
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Also, $\because \frac{d\langle \Pi \rangle}{dt} = i\hbar[\hat{\Pi}, \hat{H}]$, $\therefore$ parity remains conserved

Even parity H.O. state evolving in time
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$$|\psi\rangle\langle\psi| \equiv \begin{pmatrix} a \\ b \end{pmatrix} \otimes \begin{pmatrix} a^* & b^* \end{pmatrix} \equiv \begin{pmatrix} a \begin{pmatrix} a^* & b^* \end{pmatrix} \\ b \begin{pmatrix} a^* & b^* \end{pmatrix} \end{pmatrix} = \begin{pmatrix} aa^* & ab^* \\ ba^* & bb^* \end{pmatrix}$$

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$|0\rangle \in \mathcal{V}; \qquad |0\rangle|0\rangle \in \mathcal{V} \otimes \mathcal{V}$

$$\text{generally : } \underbrace{\mathbf{A}}_{m \times n} \otimes \underbrace{\mathbf{B}}_{p \times q} = \left[\overbrace{\begin{matrix} A_{11}B & A_{12}B & \cdots & A_{1n}B \\ A_{21}B & A_{22}B & \cdots & A_{2n}B \\ \vdots & \vdots & \vdots & \vdots \\ A_{m1}B & A_{m2}B & \cdots & A_{mn}B \end{matrix}}^{nq} \right]_{mp}$$

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and $\hat{A} = \sum_i a_i |u_i\rangle\langle u_i|$ where a_i is the eigenvalue corresponding to eigenstate $|u_i\rangle$

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Consider ortho-normal bases $\{|u_i\rangle\}$ and $\{|u'_i\rangle\}$

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Check: for spin $\frac{1}{2}$ basis:

$$\begin{pmatrix} |+\rangle \\ |-\rangle \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} |\alpha\rangle \\ |\beta\rangle \end{pmatrix}; \quad \underline{\underline{T}} \equiv \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

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for an operator $\hat{A}$,

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and $\delta(x)$ is defined as 0 everywhere else except at $x = 0$,

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Because of degeneracy, several f_i s may have the same eigenvalue k